NILESH KUMAR METAKOTA

Junior ML Engineer

nilesh.metakota@gmail.com +91 8309854539 Pithapuram, India Nilesh_Metakota

nileshmetakota17 Nilesh_Metakota

Career Objective

Aspiring Machine Learning Engineer with a strong foundation in model training, dataset preparation, and optimization. Seeking to apply academic knowledge and hands-on internship experience at Bhairav Robotics to develop intelligent automation solutions and contribute to innovative AI-driven projects. Eager to learn, adapt, and grow in a challenging environment that fosters continuous professional development.

Professional Experience

Robotics Software Engineer, *Bhairav Robotics Pvt. Ltd.* 05/2026 – Present

- Developed point-to-point navigation capabilities using the ROS 2 navigation stack.
- Implemented autonomous navigation workflows to move robots between predefined target locations.
- Integrated obstacle detection and avoidance capabilities to enable safe navigation around obstacles.
- Worked with LiDAR, GPS, and camera data for sensor fusion to support robot localization and navigation.
- Integrated navigation components to improve path planning, autonomous movement, and navigation performance.

Junior ML Engineer, *Bhairav Robotics Pvt. Ltd.* 07/2025 – 04/2026

- Collaborate with cross-functional teams to build and deploy machine learning models for robotics and automation systems.
- Prepare, preprocess, and label large datasets to enhance model accuracy and performance.
- Develop and optimize data pipelines for efficient model training, validation, and deployment workflows.
- Contribute to real-time integration of trained models in robotic environments, ensuring seamless system performance and scalability.
- Perform continuous monitoring, evaluation, and fine-tuning of models to improve precision, reliability, and response times.

Test Engineer Intern, *Bhairav Robotics Pvt. Ltd.* 03/2025 – 06/2025

• Assisted in testing and validating machine learning and automation systems.
• Worked on debugging, performance analysis, and system reliability improvements.
• Supported dataset handling and testing workflows for AI model evaluation.
• Collaborated with team members during development and deployment processes.

Full Stack Web Development Intern, *Ava Intern Edutech Pvt. Ltd.* 11/2024 – 05/2025

- Completed a project-based internship developing full-stack web applications using HTML, CSS, JavaScript, and modern frameworks.
- Designed and deployed responsive, user-friendly web interfaces, ensuring optimal user experience.
- Collaborated in agile development sprints to debug, test, and integrate new features efficiently.

- Gained practical experience in front-end and back-end architecture through real-world projects.

Projects

Fitoo – AI Integrated Meal & Workout Management System
GitHub: Fitoo Repository
- Developed **Fitoo**, a smart fitness application that combines **meal tracking, workout management, and AI-powered assistance** into a single platform.
- Integrated an **AI Assistant** to provide personalized fitness guidance, user interaction, and intelligent recommendations for workouts and nutrition.
- Designed a user-friendly and responsive interface to improve accessibility and overall user experience.
- Implemented features for tracking fitness activities, managing meal plans, and monitoring workout progress efficiently.
- Worked on backend logic, API integration, and AI-assisted automation to enhance application functionality and performance.
- Utilized technologies including **Java, AI Integration, Full Stack Development concepts, GitHub, and UI/UX Design**.
- Focused on scalable architecture, performance optimization, and real-time user interaction features.

LiDAR-Based Autonomous Navigation
GitHub: LiDAR Navigation Repository
- Developed a LiDAR-based robot navigation system using the ROS 2 navigation stack.
- Implemented point-to-point navigation to move the robot between target locations.
- Worked on obstacle detection and avoidance to enable safe autonomous navigation.
- Collaborated with the team on GPS and camera integration for navigation and environmental awareness.
- Contributed to integrating navigation components for autonomous robot movement.
- Supported navigation testing and debugging to improve robot movement and navigation reliability.

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering, 2025 – Present
Kakinada Institute of Technology and Science (KITS), AUTONOMOUS Kakinada, India

Diploma in Computer Engineering, 2022 – 2025
Aditya College of Engineering and Technology, SBTET Board
- 85.38%

Secondary Education (10th Grade), *Bhashyam EM High School, SSC Board* 2021
- 98%

Technical Skills

Programming Languages
- Python
- C++

Robotics & Navigation
- ROS 2
- Nav2
- LiDAR Navigation
- Point-to-Point Navigation
- Autonomous Navigation
- Obstacle Avoidance

Soft Skills
- Problem-Solving
- Teamwork
- Adaptability
- Communication

Machine Learning
- Model Training
- Dataset Preparation
- Evaluation
- Optimization

Web Development Tools & Platforms
- HTML
- CSS
- JavaScript
- Visual Studio Code
- Git & GitHub

Languages

Telugu
Native

English
Fluent

Hindi
Proficient

Hobbies & Interests

- Machine Learning and AI Model Development
- Exploring Emerging Technologies
- Competitive Coding and Problem-Solving
- Playing Badminton

Certifications

Full Stack Web Development Internship Certificate
- Ava Intern Edutech Pvt. Ltd.
- Certificate ID: AVA-2022-200096
- Issued on 2025-05-20